

School of Engineering, Santa Clara University

Cole Roberts

Optimizing Neural Network Resource-Efficiency for Edge Vision

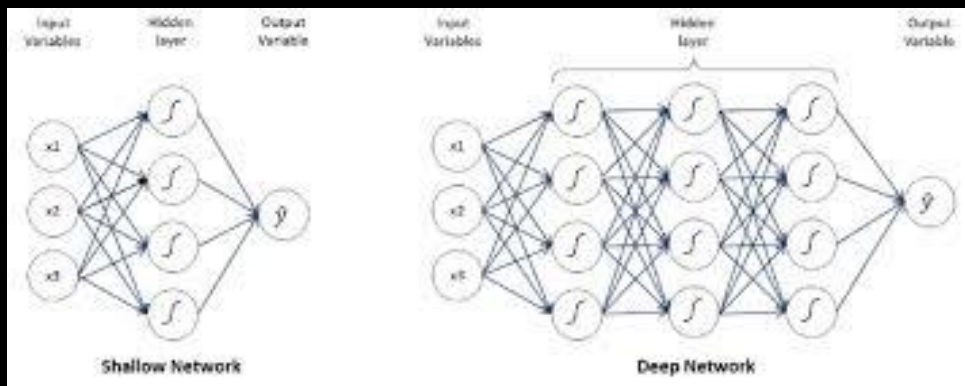
Why Fixed Computation Is Wasteful

01 Edge AI devices have strict power and latency constraints

02 Standard CNNs execute all layers for all inputs

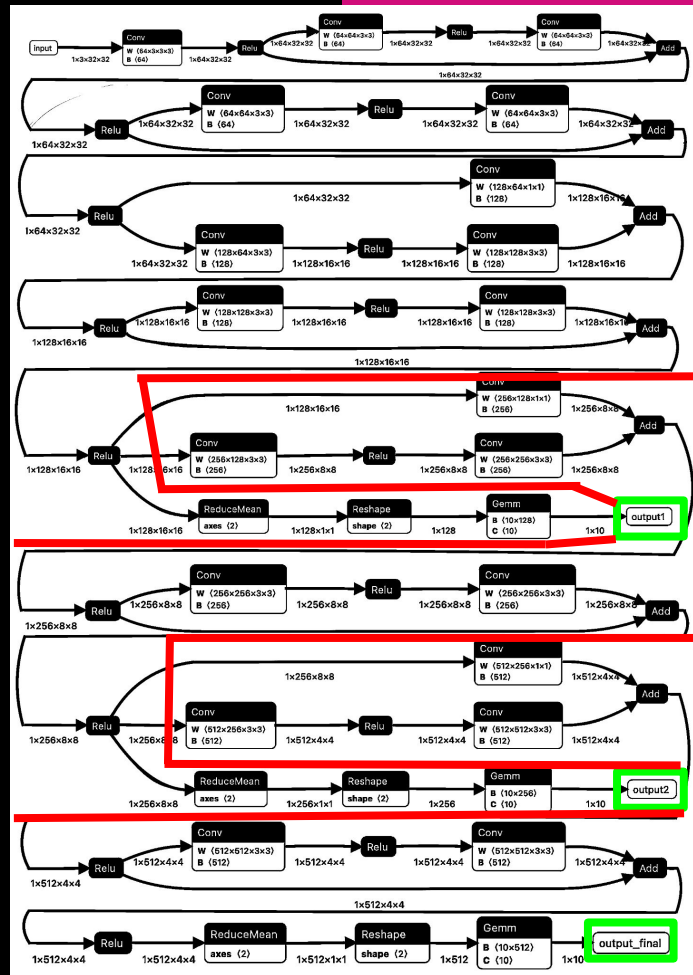
03 “Easy” images don’t need deep semantic processing

04 This leads to unnecessary compute, latency, and energy use



Adaptive Early-Exit Inference

- 01 Add intermediate classifier heads
- 02 Exit early if confidence exceeds threshold θ
- 03 Hard samples continue deeper into the network
- 04 Inspired by BranchyNet and Dynamic Neural Networks



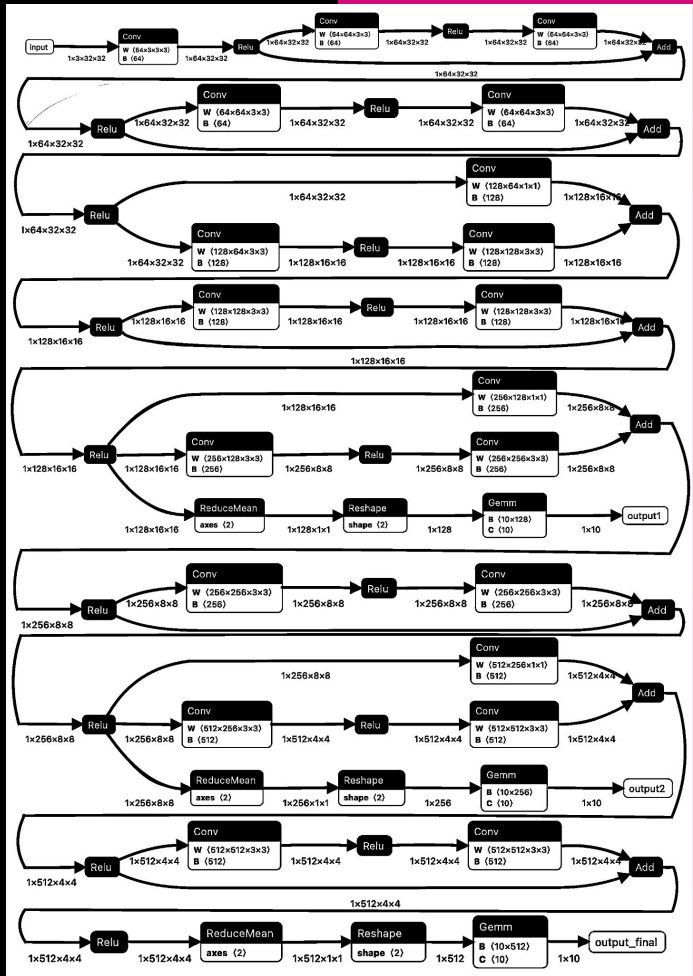
01

02

03

04

Minimal overhead



Joint Optimization

01

End-to-end training

02

Multi-exit cross-entropy loss (Equal weights for all heads)

03

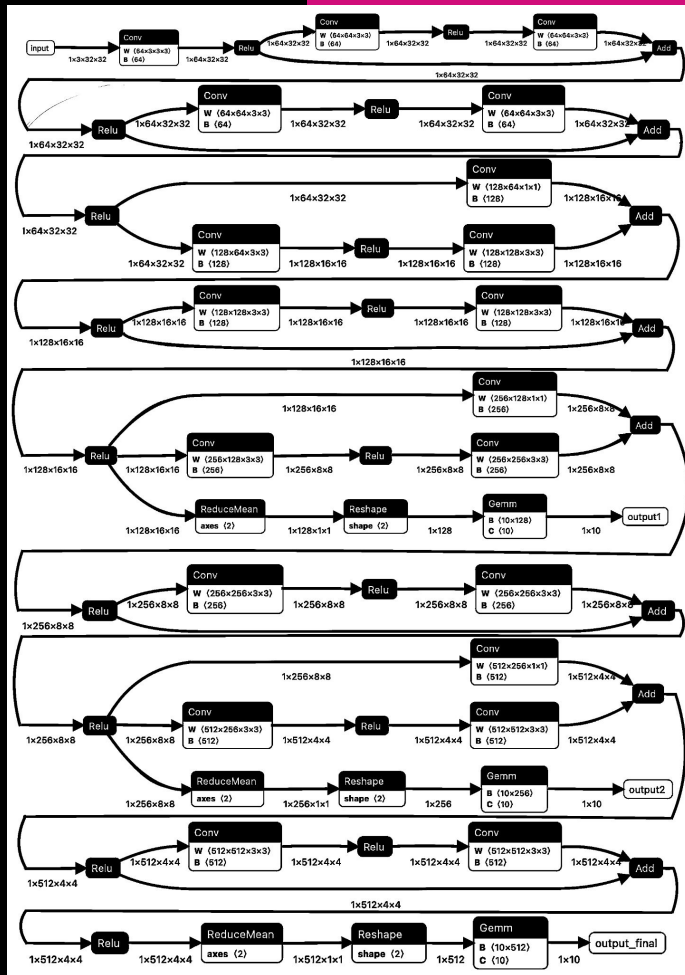
Adam optimizer

04

Cosine Annealing scheduler

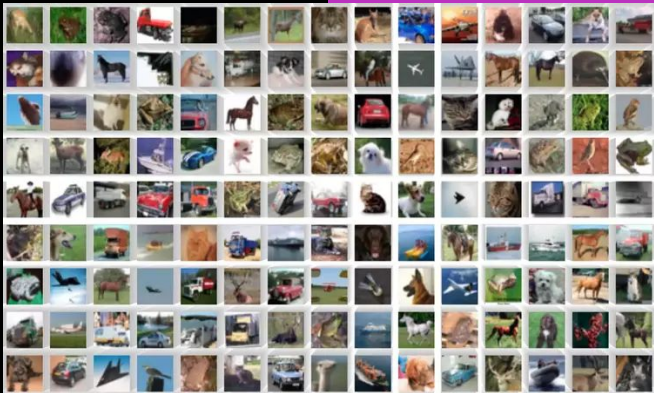
05

Data Augmentation: RandomCrop, Flip, ColorJitter



Experimental Setup

- 01 Dataset: CIFAR-10
- 02 30 Epochs
- 03 Evaluation Metrics: Top-1 Accuracy, FLOPs, Exit Distribution
- 04 Confidence thresholds: $\theta = 0.99, 0.95, 0.90, 0.80$



Accuracy and Efficiency Trade-off

Model	Top 1 Accuracy	ECE	P	Cost	FLOPs Saved	Exits (1, 2, Final)
Early Exit 1	86.39% +/- 0.28%	1.43%		288.00 MFLOPS	48.27%	10000 0 0
Early Exit 2	91.24% +/- 0.28%	2.71%		422.37 MFLOPS	24.13%	0 10000 0
Final Exit (Baseline)	91.42% +/- 0.28%	3.67%	1.00	556.71 MFLOPS	0.00%	0 0 10000
AdaptiveRes ($\theta = 0.80$)	90.74% +/- 0.29%	2.81%	0.01	331.65 EFLOPS	40.43%	7652 1447 901
AdaptiveRes ($\theta = 0.90$)	91.13% +/- 0.28%	3.23%	0.15	352.40 EFLOPS	36.70%	6718 1771 1511
AdaptiveRes ($\theta = 0.95$)	91.28% +/- 0.28%	3.41%	0.31	371.95 EFLOPS	33.19%	5848 2056 2096
AdaptiveRes ($\theta = 0.99$)	91.39% +/- 0.28%	3.64%	0.46	413.36 EFLOPS	25.75%	4068 2534 3398

Where Do Samples Exit?

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Key Insights

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Key Insights

01

Early exits recover wasted compute

02

High threshold prevents softmax overconfidence

03

θ trades accuracy for efficiency

04

Dynamic inference is ideal for edge devices

Conclusion

- 01 Adaptive inference is effective for edge vision
- 02 33% compute reduction with negligible accuracy loss
- 03 Confirms over-parameterization for average-case inputs
- 04 Learnable confidence thresholds (future work)
- 05 Deployment on real edge hardware (future work)
- 06 Energy & latency measurements (future work)